

Corporate Credit Provision

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Background: Demand-Based Asset Pricing (Kojien and Yogo, 2019)

- ▶ If demand curves for financial security inelastic, then investor-specific demand shocks contain information beyond prices.
- ▶ Demand-based asset pricing: Models that *jointly* match both prices and quantities (investors' portfolio holdings and flows)
 - ▶ Key innovation: Can trace events and patterns back to investor-level demand/supply shocks
 - ▶ Challenge: Separate demand from supply.

Back to the paper: Credit demand versus supply

- ▶ Variation in credit composition $\frac{\$ \text{Corporate Bonds}}{\$ \text{Corporate Loans}}$ across countries and time.

1. Demand side explanation: Corporate Finance

- ▶ Credit supply infinitely elastic. Investors are supplying any amount of capital at *fundamental* yield.
- ▶ Equilibrium market share of bonds versus loans entirely driven by firms' demand for different credit instrument
- ▶ Asymmetric information, Flexibility, Control etc.

Myers 1984, Leland 1994, Flannery & Rangan 2006 ...

2. Supply side explanation: Asset Pricing & Macro Finance

- ▶ Credit supply upward sloping. Investors are supplying limited amount of capital depending on yield.
- ▶ Equilibrium market share of bonds versus loans **also** driven by supply shocks interacted with elasticity of supply and demand.
- ▶ Investor leverage, investment constraints, etc.

Rajan 2006, Adrian & Shin 2014, Koijen & Yogo 2019, Bretscher et al. 2023

This paper

- ▶ Variation in $\frac{\$ \text{Corporate Bonds}}{\$ \text{Corporate Loans}}$ driven by supply side
- ▶ Structural model to **quantify** effect of macro variables (which affect bond/loan issuance via investor demand)

Discussion

- ▶ Important paper, brings asset pricing/macro finance and corporate finance closer together using granular data on investor portfolio choice and firms' balance sheets.
- ▶ Framework extends Koijen and Yogo (2019) by endogenizing corporate side. Linking investor demand to corporate decisions and **quantifying** the effects.
- ▶ Impressive data work!

Comments

1. Why nested specification?
2. Measuring investor demand
3. (Identification)

Comment 1: Why nested demand system?

- **Example.** One representative company (Apple):

1. Apple chooses whether to issue loans or bonds $l = 1, 2$:

$$\log w_t(l) = -\beta y_t(l) + \xi_t(l)$$

where $w_t(l)$ is share of l in Apple's total debt.

2. Apples bonds/loans l are held by j different investors:

$$\log w_t(j|l) = \Theta_l x_t(j) + \epsilon_{j,l}$$

- You are modeling both 1. and 2. into demand curve via nested logit.

$$L_t \sum_j (w_t(l) w_t(j|l)) = \underbrace{Q_t(l) P_t(l)}_{\text{Supply}}$$

- Why not separately model supply and demand?

- Clear differentiation between elasticity of supply and demand
- Clear differentiation supply vs demand drivers of $\frac{\$ \text{Corporate Bonds}}{\$ \text{Corporate Loans}}$

Comment 1: (Stylized) Alternative model

- Credit demand depends negatively on cost of borrowing $y_t(l)$

$$Q_t^{\text{Demand}}(l) = -\beta_D y_t(l) + \xi_t(l).$$

- Supply by investor j depends positively on yield $y_t(l)$.

$$Q_t^{j, \text{Supply}}(l) = \beta_S y_t(l) + \epsilon_t(j, l)$$

- Supply for bonds/loans is sum of j investor-specific supply curves and must equal demand. $\sum_j Q_t^{j, \text{Supply}}(l) = Q_t^{\text{Demand}}(l)$
- Equilibrium market share of credit instrument $Q_t^*(l)$ is given by

$$Q_t^*(l) = \underbrace{\xi_t(l)}_{\text{Firm Factors (Demand Side)}} + \frac{\beta_D}{\beta_S + \beta_D} \underbrace{\left(\sum_j \epsilon_t(j, l) - \xi_t(l) \right)}_{\text{Investor Factors (Supply Side)}}$$

- Role of investor-factors driven by investors willingness to substitute
- If β_S very high, then only firm factors matter (corporate finance)

Comment 2: The main regression

- ▶ Goal: Estimate how relative market share of credit type (bond vs loans) is affected by investor demand.
- ▶ The nested logit representation somewhat obfuscates the main problem: **Investor demand is unobservable!**

$$\log w_t(l) = \lambda_{lt} \left(\text{Investor Demand} \right) + \alpha - \beta y_t(l) + \varepsilon_t(l)$$

- ▶ The paper **implicitly** uses share of domestic relative to foreign investors within l as proxy for investor demand.

$$\log w_t(l) = -\lambda_{lt} \underbrace{\log w_t(0|l)}_{\text{Foreign Share}} + \alpha - \beta y_t(l) + \varepsilon_t(l)$$

- ▶ In words: Investor demand for instrument l is assumed to be high when foreign investors hold less of it.
- ▶ Could make opposite argument, especially for bonds (for which you find a negative λ !)

Comment 2: The main regression

- ▶ The authors assume that total investor demand for a credit instrument (proxied by domestic share) is not driven by yields $y_t(l)$!
- ▶ This means that the joint estimation of the demand/supply system boils down to 2 separate regressions:

1. Credit-demand regression.

$$\log w_t(l) = -\lambda_{lt} \log w_t(0|l) + \alpha - \beta_{lt} y_t(l) + \xi_t(l)$$

2. Credit-supply regression.

$$\delta_t(j, l) = \beta + \Theta_l x_t(j) + \epsilon_t(l)$$

which yields $w_t(0|l) = (1 + \sum_j \delta_t(j, l))^{-1}$

- ▶ But then: What is the difference between λ_{lt} and β_{lt} . Both measure firms' willingness to substitute between bonds and loans.

Comment 3: Brief Note on the Instrument

- ▶ Equilibrium yields y_t are driven by credit demand and supply.
- ▶ The paper assumes credit supply not driven by yields. So the focus is on identifying elasticity of credit demand.
- ▶ In order to identify credit-demand elasticity with respect to yields need **exogenous** variation in yields z_t .

Granular Instrumental Variables (Gabaix and Koijen, 2023)

- ▶ Size-weighted minus equal weighted average of firm-specific yields.
 - ▶ **Intuition:** Shocks to Apple's yield drives up aggregate yields due to Apple's size (granularity).
 - ▶ **Key assumption:** Size-weights unrelated to common factors driving yields. Should discuss this thoroughly!
- ▶ But by your logic Apple's share of bonds would decline leading to a mechanically lower aggregate $\frac{\$Bonds}{\$Loans}$ (because of Apple's size).
- ▶ Btw: Once you have z_t , which is a size-weighted sum of idiosyncratic firm-level shocks, you can use it to identify both supply and demand elasticities!

An Intruiging Finding + Potential Explanation

- ▶ Intruiging finding of the paper:
 1. Bonds: Foreign ownership and $\frac{\$Bonds}{\$Total Liabilities}$ positively related!
 2. Loans: Foreign ownership and $\frac{\$Loans}{\$Total Liabilities}$ negatively related!
- ▶ Portential explanation: Investor (supply-side) drivers dominate for bonds, firm (demand-side) dominate for loans:
 1. Foreign interest in bonds means greater aggregate investor appetite
→ firms issue more bonds!
 2. Firms prefer domestic banks for loans (less information asymmetry).
When information asymmetry decreases,
→ Firms switch to bonds and issue less loans
→ Contemporaneously foreign share rises.
- ▶ Seems like a nice example that captures the importance of investor versus firm-factors in $\frac{\$Bonds}{\$Loans}$

Conclusion

- ▶ Very rich paper. Impressive work combining so many different data sources.
- ▶ Great step towards bringing together corporate decision making and investor preferences.
- ▶ Model could potentially be refined: Separate demand from supply!
 - ▶ Differentiate elasticity of supply versus demand.
 - ▶ Separate aggregate credit supply shocks by investors and aggregate credit demand shocks by firms.
 - ▶ This allows rich decomposition of variation in bond/loan share into supply versus demand components (see appendix)

Other comments

- ▶ Does the paper rely on an incomplete passthrough from interest rates to loan rates?
 1. Loose MP $\rightarrow$ investors reach for yield $\rightarrow$ higher investor demand for corp. bonds $\rightarrow$ lower yields $\rightarrow$ corporate bond issuance increases
 2. Loose MP $\rightarrow$ Banks decrease rate on loans $\rightarrow$ loan issuance increases
- ▶ The paper argues: 1. dominates 2. and therefore share of bonds/loans increases
- ▶ How $\frac{Bonds}{Loans}$ changes wrt interest rates depends on elasticity of corporations to switch between bonds and loans.
- ▶ Direct regression of $\frac{Bonds}{Loans}$ onto r is unidentified: MP probably correlated with factors that drive choice between corp. bonds versus loans.
- ▶ Should these effects offset each other? Does the paper implicitly estimate the limited passthrough in 2. from MP to loan rates?

Framing & Semantics

- ▶ Generally, it is often unclear what the authors mean by demand and supply. Both terms are used somewhat interchangeably both in the theory and in the written exposition.
 1. Supply of corporate bonds $\Leftrightarrow$ demand for credit.
 2. Demand for corporate bonds $\Leftrightarrow$ supply of credit.
- ▶ “The supply of credit can thus be viewed as a portfolio choice problem from the perspective of the non-financial firms”.
 - ▶ I think you mean the demand for credit by firms is a portfolio choice problem. At least that is how you model it.
- ▶ “We specify the credit supply by financial sectors to depend on yields”.
 - ▶ I think you do not. In the model, demand for credit by firms (i.e. corporate bond supply) depends negatively on yields, but investors’ credit supply (i.e. corporate bond demand) does not.
- ▶ “We deviate from this literature by focusing on the determinants of credit supply rather than security (i.e. bond/loan) demand.”
 - ▶ Describe exactly how do they differ? To me they are different words for the same object.

Framing & Semantics

- ▶ Also, you currently use the following labels
 1. Firms, corporations, productive firms, non-financial firms, non-financial corporations, non-financial institutions etc.
 2. Financial institutions, financial sector, financial firms etc.
- ▶ Why don't you reduce it to **firms** versus **investors** and mention **once** that you exclude financial firms?

Alternative Model

- ▶ There is one representative firm that chooses the structure of its liabilities L_t by choosing a weight in bonds and loans $w_t(l)$ with $l = 1, 2$, and a weight in the outside asset $w_t(0)$.
- ▶ Demand is downward sloping in the yield of the instrument.

$$\log \frac{w_t(l)}{w_t(0)} = -\beta_D y_t(l) + \xi_t(l).$$

- ▶ There are $j = 1, \dots, J$ investors that provide credit across the different instruments with assets $A_{j,t}$. The investor-specific supply curve is

$$\log \frac{w_{t,j}(j)}{w_{t,j}(0)} = \beta_S y_t(l) + \epsilon_{t,j}(l).$$

For simplicity, the supply and demand elasticities are constant across investors and assets.

- ▶ Market clearing implies that

$$w_t(\textcolor{red}{y}, \xi) L_t(l) = \sum_j w_{t,j}(\textcolor{red}{y}, \epsilon) A_{t,j}$$

which has a unique fixed point $\textcolor{red}{y}$ if $\beta_D < 0$ and $\beta_S > 0$.

Alternative Specification

- ▶ Solving for yields gives

$$y_t = g_y(A_t, \xi, \epsilon, \beta_S, \beta_D)$$

- ▶ Plugging the equilibrium yield in the market share of loans/bonds:

$$w_t(l) = g_w(A_t, \xi, \epsilon, \beta_S, \beta_D)$$

- ▶ The equilibrium market share of loans versus bonds depends on the demand shocks of firms, investors, the wealth distribution across investors, and the respective elasticities of supply and demand.
- ▶ Importantly, you can run variance decompositions (see KY 2019) to see whether it is firm or investor factors that drive the majority of the variation in $w_t(l)$ over time and across countries.

Alternative Specification

- ▶ For the estimation of supply and demand elasticities you need exogenous variation in y .
- ▶ You can use the demand-based GIV to identify both supply and demand elasticities (see Gabaix and Koijen 2023)
- ▶ The main insight is that by identifying β_S you can extract the investor-specific demand shocks $\epsilon_{t,j}(l)$. This makes aggregate investor demand observable by disentangling movements along the demand curve (elasticity) from shifts in the demand curve (demand shocks).